Date: 14/08/2025

Multivariable Calculus

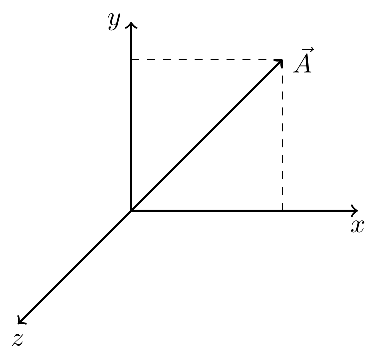
Scalar: At a given position & time, the parameter which can be specified by the magnitude only is represented as k.

Vector: At a given position & time, the parameter which can be specified by the magnitude and direction is represented as:

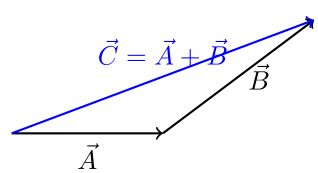
$$\vec{A} = |\vec{A}| = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

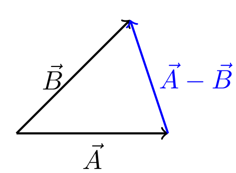
$$\text{Unit vector: } \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$\vec{A}$ = direction + magnitude $\rightarrow |\vec{A}|$



Operations





Vector Addition Obeys:

Commutative:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

Associative:

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

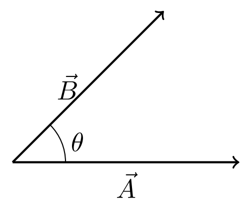
Dot Product

In cross product, the output is always a vector. Dot product output is scalar:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$

When $\theta = 90°$, $\vec{A}$ and $\vec{B}$ are uncorrelated:

$$\vec{A} \cdot \vec{B} = 0$$



Rules of Scalar Product

Commutative:

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

Distributive:

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

Vector Product or Cross Product

$$\vec{A} \times \vec{B} = \hat{n}|\vec{A}||\vec{B}| \sin\theta$$

Output will be along outward or inward of a plane.

If

$$\vec{A} = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}, \quad \vec{B} = b_x\hat{\imath} + b_y\hat{\jmath} + b_z\hat{k}$$

then

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Cross Product Rules

Distributive:

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

Commutative:

$$\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A} \quad \text{(does not hold for cross product)}$$

Associative:

$$\vec{A} \times (\vec{B} \times \vec{C}) \neq (\vec{A} \times \vec{B}) \times \vec{C}$$

Also:

$$\vec{A} \times (\vec{B} \times \vec{C}) = 0 \quad \Rightarrow \quad \hat{\imath} \times (\hat{\imath} \times \hat{\jmath}) = \hat{\imath} \times \hat{k} = 0$$

Scalar Triple Product

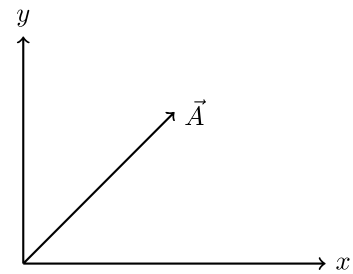
$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{C} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{C} \times \vec{A})$$

Vector Triple Product

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

(BAC–CAB rule)

Example Diagram



Differential Relation

$$ds^2 = dx^2 + dy^2$$

If:

$$x = r\cos\theta, \quad y = r\sin\theta$$

then:

$$dx = -r\sin\theta\, d\theta + \cos\theta\, dr$$
$$dy = r\cos\theta\, d\theta + \sin\theta\, dr$$

Thus:

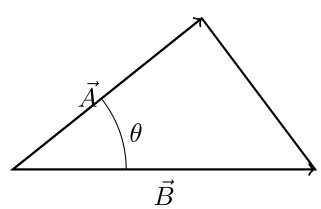
$$ds^2 = r^2\sin^2\theta\, d\theta^2 + \cos^2\theta\, dr^2 + r^2\cos^2\theta\, d\theta^2 + \sin^2\theta\, dr^2 - 2r\sin\theta\cos\theta\, dr\, d\theta + 2r\cos\theta\sin\theta\, dr\, d\theta$$

Simplifying:

$$ds^2 = r^2 d\theta^2 + dr^2$$

where r^2 is the metric coefficient.

Proof of Law of Cosines



$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

From vector form:

$$c^2 = (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B})$$
$$= A^2 + B^2 - 2|\vec{A}||\vec{B}|\cos\theta$$

If $\theta = \pi - \alpha$:

$$c^2 = A^2 + B^2 - 2AB\cos\alpha$$

Thus:

$$c = \sqrt{A^2 + B^2 - 2AB\cos\alpha}$$

Coordinate Components

If $\vec{A}$ makes angle α with the x-axis:

$$A_y = A\cos\theta, \quad A_z = A\sin\theta$$

When rotated by angle α:

$$A'_y = A\cos(\theta+\alpha), \quad A'_z = A\sin(\theta+\alpha)$$

Using addition formulas:

$$A'_y = A\cos\theta\cos\alpha - A\sin\theta\sin\alpha$$
$$A'_z = A\sin\theta\cos\alpha + A\cos\theta\sin\alpha$$

Matrix form:

$$\begin{bmatrix} A'_y \\ A'_z \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} A_y \\ A_z \end{bmatrix}$$

Differentials

We need to express the differential change in length corresponding to differential change in coordinate system:

$$dl^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2$$

where h_i are metric coefficients.

Polar Coordinates

Transformation:

$$(x, y) \to (r, \theta), \quad u_1 = r, \quad u_2 = \theta$$

Metric coefficients:

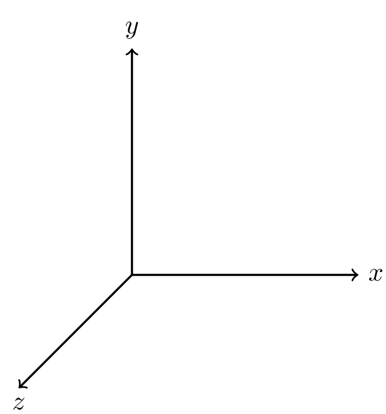
$$dl_r = h_1 dr, \quad h_1 = 1$$
$$dl_\theta = h_2 d\theta, \quad h_2 = r$$

Therefore:

$$\vec{dl} = \hat{a}_r dr + \hat{a}_\theta (r\, d\theta)$$
$$dl^2 = (dr)^2 + (r\, d\theta)^2$$

Implemented only in the case of an orthogonal coordinate system.

Rectangular Coordinate System



For rectangular coordinates:

$$(u_1, u_2, u_3) \to (x, y, z), \quad (h_1, h_2, h_3) = (1, 1, 1)$$

Differential length element:

$$d\vec{l} = \hat{a}_x\, dx + \hat{a}_y\, dy + \hat{a}_z\, dz$$

Magnitude:

$$|dl| = \sqrt{dx^2 + dy^2 + dz^2}$$

Generalized Differential Formulas

For a general orthogonal coordinate system:

$$d\vec{l} = h_1\hat{a}_1\, du_1 + h_2\hat{a}_2\, du_2 + h_3\hat{a}_3\, du_3$$

Differential volume:

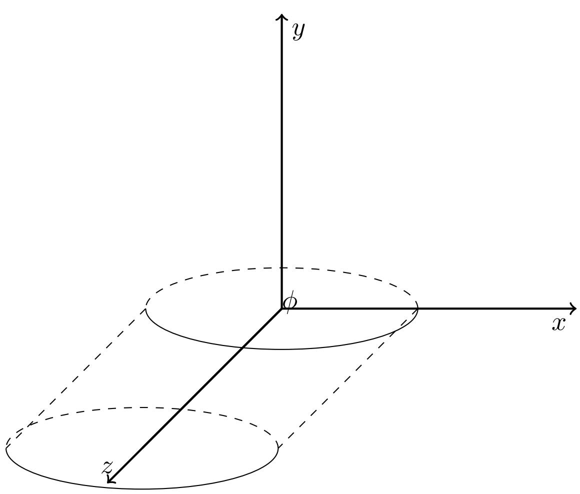
$$dv = h_1h_2h_3\, du_1\, du_2\, du_3$$

Differential Change in Area

For planes perpendicular to coordinate axes:

$$\begin{aligned} dS_x &= dy\, dz, \quad dS_x = h_2h_3\, du_2\, du_3 \\ dS_y &= dz\, dx, \quad dS_y = h_3h_1\, du_3\, du_1 \\ dS_z &= dx\, dy, \quad dS_z = h_1h_2\, du_1\, du_2 \end{aligned}$$

Cylindrical Coordinate System



Parametric relations:

$$x = r\cos\phi, \quad y = r\sin\phi, \quad z = z$$

Coordinates:

$$(u_1, u_2, u_3) = (r, \phi, z)$$

Metric coefficients:

$$h_1 = 1, \quad h_2 = r, \quad h_3 = 1$$

Differential length vector:

$$d\vec{l} = \hat{a}_r\, dr + \hat{a}_\phi\, r\, d\phi + \hat{a}_z\, dz$$

Magnitude:

$$|dl| = \sqrt{dr^2 + r^2\, d\phi^2 + dz^2}$$

Differential volume:

$$dv = r\, dr\, d\phi\, dz$$

Differential area elements:

$$dS_r = r\, d\phi\, dz$$

$$dS_\phi = dr\, dz$$

$$dS_z = r\, dr\, d\phi$$

Spherical Coordinate System

Coordinate Transformation

$$\begin{aligned} x &= R\sin\theta\cos\phi \\ y &= R\sin\theta\sin\phi \\ z &= R\cos\theta \end{aligned}$$

General Form of a Sphere

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$$

For origin-centered sphere:

$$x^2 + y^2 + z^2 = R^2$$

Differential Displacement Vector

$$d\vec{l} = \hat{a}_R\, dR + \hat{a}_\theta\, R\, d\theta + \hat{a}_\phi\, R\sin\theta\, d\phi$$

Volume Element

$$dV = R^2 \sin\theta\, dR\, d\theta\, d\phi$$

Parametric Equations

$$\begin{aligned} x &= R\sin\theta\cos\phi \\ y &= R\sin\theta\sin\phi \\ z &= R\cos\theta \end{aligned}$$

$$R = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right), \quad \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Surface Elements

$$\begin{aligned} ds_R &= R^2 \sin\theta\, d\theta\, d\phi \\ ds_\theta &= R\sin\theta\, dR\, d\phi \\ ds_\phi &= R\, dR\, d\theta \end{aligned}$$

Conversion from Cylindrical to Cartesian Coordinates

From cylindrical coordinates (ρ, ϕ, z) to Cartesian (x, y, z):

$$\begin{aligned} x &= \rho \cos\phi \\ y &= \rho \sin\phi \\ z &= z \end{aligned}$$

Vector transformation:

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y$$

with:

$$\begin{aligned} A_x &= A_\rho \cos\phi - A_\phi \sin\phi \\ A_y &= A_\rho \sin\phi + A_\phi \cos\phi \end{aligned}$$

$$A_y = \vec{A} \cdot \hat{a}'_y$$

$$A_y = A_x \hat{a}_x \cdot \hat{a}'_y + A_y \hat{a}_y \cdot \hat{a}'_y$$

Using dot products:

$$\begin{aligned} A_x &= A'_x \cos\phi - A'_y \sin\phi \\ A_y &= A'_x \sin\phi + A'_y \cos\phi \\ A_z &= A'_z \end{aligned}$$

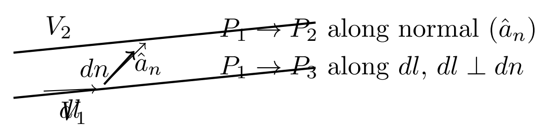
In matrix form:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A'_x \\ A'_y \\ A'_z \end{bmatrix}$$

Date: 20/08/2025

Gradient of a scalar function:

Let us consider a scalar fn $V(u_1, u_2, u_3)$.



Space rate of change dV is greatest along dn because dn is the shortest distance between two points in planes V_1, V_2.

$$\nabla V = \text{grad}(V) = \hat{a}_n \frac{dV}{dn}$$

$$\frac{dV}{dl} = \frac{dV}{dn} \cdot \frac{dn}{dl} = \frac{dV}{dn} \cos\theta = \frac{dV}{dn} (\hat{a}_n \cdot \hat{a}_l) = \nabla V \cdot \hat{a}_l$$

$$dV = \frac{\partial V}{\partial u_1} du_1 + \frac{\partial V}{\partial u_2} du_2 + \frac{\partial V}{\partial u_3} du_3$$

$$= \left(\frac{\partial V}{\partial u_1} \hat{a}_{u_1} + \frac{\partial V}{\partial u_2} \hat{a}_{u_2} + \frac{\partial V}{\partial u_3} \hat{a}_{u_3} \right) \cdot \left(\hat{a}_{u_1} du_1 + \hat{a}_{u_2} du_2 + \hat{a}_{u_3} du_3 \right) = \nabla V \cdot d\ell$$

Gradient in C.S

$$\nabla V = \hat{a}_{u_1} \frac{1}{h_1}\frac{\partial V}{\partial u_1} + \hat{a}_{u_2} \frac{1}{h_2}\frac{\partial V}{\partial u_2} + \hat{a}_{u_3} \frac{1}{h_3}\frac{\partial V}{\partial u_3}$$

R.C.S

$$(u_1, u_2, u_3) \to (x, y, z), \qquad (h_1, h_2, h_3) \to (1, 1, 1)$$
$$\nabla V = \hat{a}_x \frac{\partial V}{\partial x} + \hat{a}_y \frac{\partial V}{\partial y} + \hat{a}_z \frac{\partial V}{\partial z}$$

Cylindrical C.S

$$(u_1, u_2, u_3) \to (\rho, \phi, z), \qquad (h_1, h_2, h_3) \to (1, \rho, 1)$$
$$\nabla V = \hat{a}_\rho \frac{\partial V}{\partial \rho} + \hat{a}_\phi \frac{1}{\rho}\frac{\partial V}{\partial \phi} + \hat{a}_z \frac{\partial V}{\partial z}$$

Spherical C.S

$$(u_1, u_2, u_3) \to (r, \theta, \phi), \qquad (h_1, h_2, h_3) \to (1, r, r\sin\theta)$$
$$\nabla V = \hat{a}_r \frac{\partial V}{\partial r} + \hat{a}_\theta \frac{1}{r}\frac{\partial V}{\partial \theta} + \hat{a}_\phi \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}$$

Example & Gradient of vector function (excluded from syllabus)

(a) $V(x, y, z) = V_0\, e^{-x}\, \sin\Big(\frac{\pi y}{4}\Big)$

$$\nabla V = \hat{a}_x \frac{\partial}{\partial x}\big(V_0 e^{-x} \sin \tfrac{\pi y}{4}\big) + \hat{a}_y \frac{\partial}{\partial y}\big(V_0 e^{-x} \sin \tfrac{\pi y}{4}\big)$$
$$= \hat{a}_x(-V_0 e^{-x} \sin \tfrac{\pi y}{4}) + \hat{a}_y \Big(V_0 e^{-x} \cdot \tfrac{\pi}{4} \cos \tfrac{\pi y}{4}\Big)$$

Gradient of vector function (excluded from syllabus)

$$\vec{A} = \hat{a}_x A_x + \hat{a}_y A_y + \hat{a}_z A_z$$

$$\nabla \vec{A} = \big(\hat{a}_x \frac{\partial}{\partial x} + \hat{a}_y \frac{\partial}{\partial y} + \hat{a}_z \frac{\partial}{\partial z}\big)\big(\hat{a}_x A_x + \hat{a}_y A_y + \hat{a}_z A_z\big)$$
$$= \begin{bmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{bmatrix} \quad \text{(dyadic form)}$$

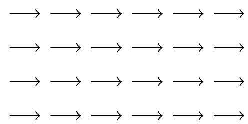
Divergence: (of vector function) Scalar fn $\longleftarrow \nabla \cdot \vec{A} \longrightarrow$ Vector fn.

Divergence of a vector fn at a point can be defined as *net outward flux/field lines/vector of $\vec{A}$ per unit volume as volume about that point tends to zero.*

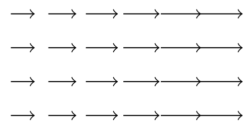
Divergence of a vector function — sketches

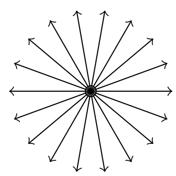
$$\nabla V = \hat{a}_x(-V_0 e^{-x} \sin \tfrac{\pi y}{4}) + \hat{a}_y \Big(\tfrac{\pi}{4} V_0 e^{-x} \cos \tfrac{\pi y}{4}\Big)$$

Field sketches



divergence = 0

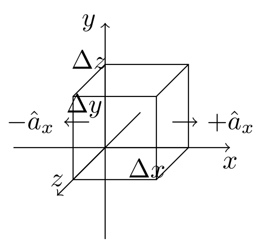




Divergence (definition by flux limit)

Divergence of a vector at a point can be defined as net outward flux/field lines/vector of $\vec{A}$ per unit volume as volume about the point tends to zero:

$$\boxed{\operatorname{div}\vec{A} = \lim_{\Delta V \to 0} \frac{\oint \vec{A}\cdot\hat{n}\,dS}{\Delta V}}$$



Flux of $\vec{A}$ at point (x_0, y_0, z_0) over enclosed surface:

$$\oint_{\text{enclosed surf}} \vec{A}\cdot d\vec{S} = \int_{\text{front face}} + \int_{\text{top face}} + \int_{\text{back face}} + \int_{\text{bottom face}} + \int_{\text{left face}} + \int_{\text{right face}}$$

Front & back faces (Taylor expansion)

$$A_{\text{front face}} = A_x\left(x_0 + \frac{\Delta x}{2},\, y_0,\, z_0\right)\,\Delta y\,\Delta z \quad \text{(Expanded using Taylor's theorem)}$$

$$= \left[A_x(x_0,y_0,z_0) + \frac{\Delta x}{2}\frac{\partial A_x}{\partial x}\bigg|_{x_0,y_0,z_0} + \text{H.O.T.}\right]\Delta y\,\Delta z$$

$$\oint \vec{A}\cdot d\vec{S} = A_{\text{back face}} + \Delta\Phi_{\text{back face}} \qquad \text{(back-face normal } = -\hat{a}_x\text{)}$$

$$A_{\text{back face}} = A_x\left(x_0 - \frac{\Delta x}{2},\, y_0,\, z_0\right)\,(-\hat{a}_x)\,\Delta y\,\Delta z$$

$$= \left[A_x(x_0,y_0,z_0) - \frac{\Delta x}{2}\frac{\partial A_x}{\partial x}\bigg|_{x_0,y_0,z_0} + \text{H.O.T.}\right]\Delta y\,\Delta z$$

Sum over all 6 faces ⇒ divergence

$$\oint_{(\text{front+back+top+bottom+left+right})} \vec{A} \cdot d\vec{S} = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) \Delta x \, \Delta y \, \Delta z$$

$$\therefore \quad \text{div}\,\vec{A} = \lim_{\Delta V \to 0} \frac{\Phi_A}{\Delta V} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \vec{A} = \left(\hat{a}_x \frac{\partial}{\partial x} + \hat{a}_y \frac{\partial}{\partial y} + \hat{a}_z \frac{\partial}{\partial z}\right) \cdot (\hat{a}_x A_x + \hat{a}_y A_y + \hat{a}_z A_z) = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\boxed{\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(h_2 h_3 A_1) + \frac{\partial}{\partial u_2}(h_1 h_3 A_2) + \frac{\partial}{\partial u_3}(h_1 h_2 A_3)\right]} \quad \text{(gen. orthogonal C.S)}$$

Rectangular Coordinate System

$$(h_1, h_2, h_3) = (1,\ 1,\ 1)$$
$$(u_1, u_2, u_3) = (x,\ y,\ z)$$
$$A_1,\ A_2,\ A_3 = A_x,\ A_y,\ A_z$$

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Cylindrical Coordinate System

$$(h_1, h_2, h_3) = (1,\ r,\ 1)$$
$$(u_1, u_2, u_3) = (r,\ \phi,\ z)$$
$$A_1,\ A_2,\ A_3 = A_r,\ A_\phi,\ A_z$$

$$\nabla \cdot \vec{A} = \frac{1}{r}\left[\frac{\partial}{\partial r}(r A_r) + \frac{\partial}{\partial \phi}(A_\phi) + \frac{\partial}{\partial z}(r A_z)\right]$$

Divergence in spherical C.S

Using $(h_1, h_2, h_3) = (1,\ r,\ r\sin\theta)$ and $(u_1, u_2, u_3) = (r, \theta, \phi)$:

$$\boxed{\nabla \cdot \vec{A} = \frac{1}{r^2 \sin\theta}\left[\frac{\partial}{\partial r}\big(r^2 \sin\theta\, A_r\big) + \frac{\partial}{\partial \theta}\big(r \sin\theta\, A_\theta\big) + \frac{\partial}{\partial \phi}\big(r\, A_\phi\big)\right]}$$

which reduces to the standard form

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial \phi}.$$

If there is no source, then divergence $= 0$.

- No charge in magnetic field ⇒ $\nabla \cdot \vec{H} = 0$.
- Magnetic field created by current carrying conductor (fundamental of electric/magnetic field).
- Gauss's law: $\nabla \cdot \vec{E} = \rho/\varepsilon$ (charge density; often $\varepsilon_0 = 1$ in normalized units).
- No charge is in enclosed sphere ⇒ $\nabla \cdot \vec{E} = 0$.

Solenoidal Functions

Vector functions for which divergence = 0, they are known as **Solenoidal Functions**.

$$\vec{B} = a_\phi \left(\frac{k}{r}\right) \quad \text{(Cylindrical coordinate system)}$$

$$\nabla \cdot \vec{B} = 0$$

Since only B_ϕ is present, so only $\frac{\partial B_\phi}{\partial \phi}$ exists:

$$\nabla \cdot \vec{B} = \frac{1}{r}\left[\frac{\partial}{\partial \phi}(B_\phi)\right]$$

$$\nabla \cdot \vec{B} = \frac{1}{r}\left[\frac{\partial}{\partial \phi}\left(\frac{k}{r}\right)\right] = 0$$

where $\frac{\partial}{\partial \phi}\left(\frac{k}{r}\right) = 0$.

Solved Problems

Q1.

$$A = a_x x^2 + a_y\, xy + a_z\, yz$$

$$\nabla \cdot A = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot A = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(yz)$$

$$\nabla \cdot A = 2x + y + 0 = 2x + y$$

$$\therefore \nabla \cdot A = 2x + y$$

Q2.

$$A = a_x\left(\frac{1}{x}\right) + a_y\left(\frac{y}{z}\right) + a_z\left(\frac{x}{y}\right)$$

$$\nabla \cdot A = \frac{\partial}{\partial x}\left(\frac{1}{x}\right) + \frac{\partial}{\partial y}\left(\frac{y}{z}\right) + \frac{\partial}{\partial z}\left(\frac{x}{y}\right)$$

$$\nabla \cdot A = -\frac{1}{x^2} + \frac{1}{z} + 0$$

$$\therefore \nabla \cdot A = \frac{1}{z} - \frac{1}{x^2}$$

Curl

$$\oint (\nabla \times \vec{F})\, d\ell$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

You have to take integration for closed paths.

The total sum of this would be $(\nabla \times \vec{F})$

Date: 25/08/2025

Divergence Theorem

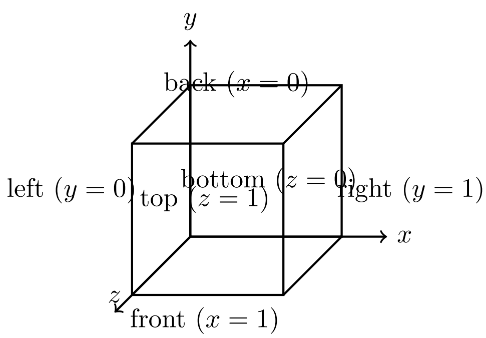
The volume integral of divergence of a vector $\vec{A}$ is equal to the net outward flow of the vector through the surface that bounds the volume.

$$\int_V (\nabla \cdot \vec{A})\, dv = \oint_S \vec{A} \cdot d\vec{S}$$

—

Example: Cube Verification

Given $\vec{A} = \hat{a}_x x^2 + \hat{a}_y xy + \hat{a}_z yz$, verify Divergence Theorem over a cube of unit length with one corner at the origin in the first octant.



—

Flux Calculation

Front face ($x = 1$):

$$\iint A_x\, dy\, dz = \iint x^2\, dy\, dz = 1$$

Back face ($x = 0$):

$$\iint 0\, dy\, dz = 0$$

$$\Rightarrow \text{Front} + \text{Back} = 1$$

Right face ($y = 1$):

$$\iint xy\, dx\, dz = \frac{1}{2}$$

Left face ($y = 0$):

$$\iint 0\, dx\, dz = 0$$

$$\Rightarrow \text{Right} + \text{Left} = \tfrac{1}{2}$$

Top face ($z = 1$):

$$\iint yz\, dx\, dy = \frac{1}{2}$$

Bottom face ($z = 0$):

$$\iint 0\, dx\, dy = 0$$

$$\Rightarrow \text{Top} + \text{Bottom} = \tfrac{1}{2}$$

Total Flux

$$\oint \vec{A} \cdot d\vec{S} = 1 + \tfrac{1}{2} + \tfrac{1}{2} = 2$$

—

Volume Integral

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(yz) = 2x + x + y = 3x + y$$

$$\iiint (3x + y)\, dx\, dy\, dz = 2$$

Hence, Divergence Theorem is verified.
—

Curl of a Vector

Circulation of a vector around a contour C is:

$$\oint \vec{A} \cdot d\vec{l}$$

Curl of $\vec{A}$:

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Generalized Orthogonal Coordinates

$$\nabla \times \vec{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{a}_1 h_1 & \hat{a}_2 h_2 & \hat{a}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

Cylindrical Coordinates

$h_1 = 1, h_2 = \rho, h_3 = 1$ with $u_1 = \rho, u_2 = \phi, u_3 = z$:

$$\nabla \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} \hat{a}_\rho \rho & \hat{a}_\phi & \hat{a}_z \\ \dfrac{\partial}{\partial \rho} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Spherical Coordinates

$h_1 = 1, h_2 = R, h_3 = R \sin\theta$:

$$\nabla \times \vec{A} = \frac{1}{R^2 \sin\theta} \begin{vmatrix} \hat{a}_R R^2 \sin\theta & \hat{a}_\theta R \sin\theta & \hat{a}_\phi \\ \dfrac{\partial}{\partial R} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ A_R & R A_\theta & R \sin\theta A_\phi \end{vmatrix}$$

—

Derivation of $(\nabla \times \vec{A})_x$

$$(\nabla \times A)_x = \lim_{\Delta y \Delta z \to 0} \frac{1}{\Delta y \Delta z} \oint A \cdot dl$$

$$\oint A \cdot dl = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \Delta y \Delta z$$

Hence,

$$(\nabla \times A)_x = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right)$$

—

Side-1

$$dl = \hat{a}_z \Delta z$$

$$A \cdot dl = A_z(x_0, y_0 + \Delta y, z_0) \Delta z$$

Using Taylor expansion:

$$A_z(x_0, y_0 + \Delta y, z_0) = A_z(x_0, y_0, z_0) + \frac{\Delta y}{2} \frac{\partial A_z}{\partial y} \bigg|_{x_0, y_0, z_0} + \text{H.O.T.}$$

$$A \cdot dl = \left[A_z(x_0, y_0, z_0) + \frac{\Delta y}{2} \frac{\partial A_z}{\partial y} + \text{H.O.T.} \right] \Delta z$$

—

Side-3

$$dl = -\hat{a}_z \Delta z$$

$$A \cdot dl = -\left[A_z(x_0, y_0, z_0) + \frac{\Delta y}{2} \frac{\partial A_z}{\partial y} + \text{H.O.T.} \right] \Delta z$$

$$= -A_z(x_0, y_0, z_0) \Delta z - \frac{\Delta y}{2} \frac{\partial A_z}{\partial y} \Delta z + \text{H.O.T.}$$

By combining sides 1 and 3:

$$\Delta = \frac{\partial A_z}{\partial y} \Delta y \Delta z$$

—

Side-2

$$dl = -\hat{a}_y \Delta y$$
$$A \cdot dl = -A_y(x_0, y_0, z_0 + \Delta z)\Delta y$$

Side-4

$$dl = \hat{a}_y \Delta y$$
$$A \cdot dl = A_y(x_0, y_0, z_0)\Delta y$$

By combining sides 2 and 4 we get:

$$\Delta = -\frac{\partial A_y}{\partial z}\Delta z \Delta y$$

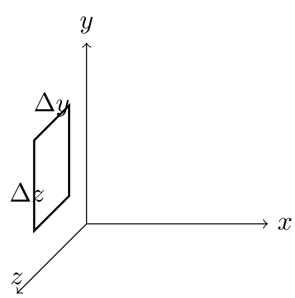
—

Final Expression

$$\oint A \cdot dl = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\Delta y \Delta z$$
$$(\nabla \times A)_x = \lim_{\Delta y \Delta z \to 0} \frac{1}{\Delta y \Delta z}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\Delta y \Delta z$$
$$(\nabla \times A)_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}$$

—

Illustration



Stokes Theorem (02/09/25)

Stokes Theorem: The surface integral of a curl of a vector over an open surface S is equal to the closed line integral of the vector along the contour bounding the surface:

$$\iint_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_{\partial S} \vec{A} \cdot d\vec{l}$$

Finding $\nabla \times \vec{A}$ in cylindrical coordinates

Given

$$\vec{A} = A_\phi \, \hat{a}_\phi$$

In general:

$$\vec{A} = A_r \hat{a}_r + A_\phi \hat{a}_\phi + A_z \hat{a}_z$$

$$\nabla \times \vec{A} = \lim_{\Delta v \to 0} \frac{1}{\Delta v} \oint \vec{A} \cdot \Delta \vec{l} = 0 \quad (\text{since } A_r = A_z = 0)$$

Using determinant form:

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Since $A_y = A_z = 0$:

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & A_\phi & 0 \end{vmatrix} = 0$$

Thus, $\vec{A}$ is an **irrotational vector function**.

Scalar Function V

Gradient:

$$\nabla V$$

Divergence of gradient:

$$\nabla \cdot \nabla V = \nabla^2 V$$

In Cartesian coordinates:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Hence,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2 V = 0 \quad \Rightarrow \quad \text{Laplace equation}$$

Cylindrical System

$$\left(\hat{a}_r \frac{\partial}{\partial r} + \hat{a}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{a}_z \frac{\partial}{\partial z} \right) \cdot \left(\hat{a}_r \frac{\partial V}{\partial r} + \hat{a}_\phi \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{a}_z \frac{\partial V}{\partial z} \right)$$

$$= \frac{\partial^2 V}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Thus,

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2 V = k \quad \Rightarrow \quad \text{Poisson equation}$$

Laplace Equation Example

We start with:

$$\nabla^2 V = 0$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Since only variation in z:

$$0 + 0 + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{\partial}{\partial z}\left(\frac{\partial V}{\partial z}\right) = 0 \quad \Rightarrow \quad \frac{\partial V}{\partial z} = A$$

Thus:

$$V = Az + B$$

Boundary Conditions

At $z = 0$:

$$V(0) = 0 \quad \Rightarrow \quad B = 0$$

At $z = d$:

$$V(d) = V_0 \quad \Rightarrow \quad A = \frac{V_0}{d}$$

Hence:

$$V(z) = \frac{V_0}{d} z$$

Electric Field Intensity

$$\vec{E} = -\nabla V$$

$$= -\left(\hat{a}_x \frac{\partial V}{\partial x} + \hat{a}_y \frac{\partial V}{\partial y} + \hat{a}_z \frac{\partial V}{\partial z}\right)$$

$$= -\hat{a}_z \frac{\partial V}{\partial z}$$

Since $\frac{\partial V}{\partial z} = \frac{V_0}{d}$:

$$\vec{E} = -\hat{a}_z \frac{V_0}{d}$$

Identity: Scalar Function

$$\nabla \times \nabla V = 0$$

$$\nabla \times \nabla V = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \hat{a}_x \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y}\right) + \hat{a}_y \left(\frac{\partial^2 V}{\partial z \partial x} - \frac{\partial^2 V}{\partial x \partial z}\right) + \hat{a}_z \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x}\right) = 0$$

Using Stokes Theorem

$$\iint (\nabla \times \vec{A}) \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{l}$$

$$\oint \nabla V \cdot d\vec{l} = \text{directional derivative}$$

$$= \iint \nabla V \cdot d\vec{S} = 0 \quad \text{(over closed path)}$$

Curl of Curl

$$\nabla \times (\nabla \times \vec{A}) = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Expansion:

$$= \hat{a}_x \left(\frac{\partial}{\partial y} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right) + \dots$$

After simplification:

$$\nabla \times (\nabla \times \vec{A}) = 0$$

Flux through a Cylinder

Given:

$$\vec{F} = \hat{a}_x \frac{k_1}{r} + \hat{a}_z \frac{k_2 z}{r}$$

We compute flux over closed cylinder surface defined by $r = 2$, $z = \pm 3$:

$$\iint \vec{F} \cdot d\vec{s} = \iint_{\text{top}} \vec{F} \cdot d\vec{s} + \iint_{\text{side}} \vec{F} \cdot d\vec{s} + \iint_{\text{bottom}} \vec{F} \cdot d\vec{s}$$

Top surface ($z = 3$):

$$d\vec{s} = \hat{a}_z \, r \, dr \, d\phi$$

$$\vec{F} \cdot d\vec{s} = \left(\hat{a}_x \frac{k_1}{r} + \hat{a}_z \frac{k_2 z}{r} \right) \cdot (\hat{a}_z \, r \, dr \, d\phi)$$

$$= k_2 z \, r \, dr \, d\phi \quad \Big|_{z=3}$$

$$= 3k_2 \, r \, dr \, d\phi$$

$$\oint \vec{F} \cdot \hat{n} \, ds = \int_0^{2\pi} \int_0^R 3k_z r \, dr \, d\phi$$

$$= 3k_z \int_0^R r^2 \, dr \int_0^{2\pi} d\phi = 3k_z \left(\frac{R^2}{2} \right) (2\pi) = 12\pi k_z$$

Bottom surface $z = -3$:

$$\oint \vec{F} \cdot \hat{n} \, ds = -12\pi k_z$$

Side wall $r = 6$:

$$\hat{n} \, ds = \hat{a}_r \, r \, dz \, d\phi$$

$$\oint \vec{F} \cdot \hat{n}\, ds = (\hat{a}_x k_1 + \hat{a}_z k_2) \cdot (\hat{a}_r\, r\, d\phi\, dz)$$

$$= \frac{k_1}{r}(r\, dz\, d\phi) + \iint (k_1\, dz\, d\phi + k_2\, r\, dz\, d\phi)$$

$$= k_1 \int_0^3 dz \int_0^{2\pi} d\phi + k_2 \int_0^3 dz \int_0^{2\pi} 6\, d\phi = 24\pi k_1 + 12\pi k_2$$

$$= 12\pi(2k_1 + k_2)$$

—

Q) Evaluate $\int \vec{F} \cdot d\vec{r}$ along quarter circle $x^2 + y^2 = 9$

Equation: $y = \pm\sqrt{9 - x^2}$, radius $= 3$

$$\vec{F} = (2x, y, -2x)$$
$$d\vec{r} = \hat{a}_x dx + \hat{a}_y dy$$
$$\vec{F} \cdot d\vec{r} = (2x, y, -2x) \cdot (\hat{a}_x dx + \hat{a}_y dy) = 2xdx + ydy$$

Therefore,

$$\int \vec{F} \cdot d\vec{r} = \int (2xdx + ydy)$$

Using $x = \sqrt{9 - y^2}$,

$$\int_0^3 x\, dy - 2\int_0^3 \sqrt{9 - y^2}\, dy$$
$$= 9\left(1 + \frac{\pi}{2}\right)(-1)$$

—

Vector Line Integral

Let $\vec{r}(t) = g(t)\hat{i} + h(t)\hat{j} + k(t)\hat{k}, \quad a \le t \le b$

$$S_n = \sum f(x_m, y_m, z_m)\Delta S_m$$

Integral of f over curve a to b:

$$\int f(x, y, z)\, ds$$

If $\vec{r}(t)$ is smooth:

$$v(t) = \frac{d\vec{r}(t)}{dt}, \quad \rho(t) = v(t)\, dt$$

Arc length:

$$S(t) = \int_a^t |\vec{v}(\tau)| d\tau$$

Therefore,

$$\int_C f(x, y, z)\, ds = \int_a^b f(g(t), h(t), k(t))\, |\vec{v}(t)|\, dt$$

—

Example

Integrate $f(x, y, z) = x - 3y^2 + z$ over line segment joining $(0, 0, 0)$ to $(1, 1, 1)$.

$$\vec{r}(t) = g(t)\hat{\imath} + h(t)\hat{\jmath} + k(t)\hat{k}, \quad g(t) = t, h(t) = t, k(t) = t, \quad 0 \le t \le 1$$

$$\vec{v}(t) = \frac{d}{dt}(t\hat{\imath} + t\hat{\jmath} + t\hat{k}) = (1, 1, 1), \quad |\vec{v}(t)| = \sqrt{3}$$

$$\int_C f(x, y, z)\, ds = \int_0^1 (t - 3t^2 + t)\sqrt{3}\, dt$$

$$= \sqrt{3}\int_0^1 (2t - 3t^2)\, dt = \sqrt{3}\left[t^2 - t^3\right]_0^1 = \sqrt{3}(1 - 1) = 0$$

—

Cone Equation in Cylindrical Coordinates

$$x = R\cos\theta, \quad y = R\sin\theta, \quad z = R$$

$$x^2 + y^2 = R^2, \quad \sqrt{x^2 + y^2} = R = z$$

Flux across the plane curve (11/09/25)

$$\oint F \cdot dr = \oint (M\, dy - N\, dx)$$

Example

Find flux of

$$F = (x - y)\hat{\imath} + xy\hat{\jmath}$$

across the circle $x^2 + y^2 = 1$ in the xy-plane.

$$M = (x - y), \quad N = xy, \quad F = M\hat{\imath} + N\hat{\jmath}$$
$$x = \cos t, \quad y = \sin t$$
$$r(t) = \cos t\, \hat{\imath} + \sin t\, \hat{\jmath}$$
$$M = \cos t - \sin t, \quad N = \sin t \cos t$$

$$\text{Flux} = \oint (M\, dy - N\, dx)$$

$$= \oint \big((\cos t - \sin t)(\cos t dt) - (\cos t \sin t dt)\big)$$

$$= \oint (\cos^2 t - \sin t \cos t + \sin t \cos t - \cos t \sin t)dt$$

$$= \oint \cos^2 t\, dt$$

$$= \int_0^{2\pi} \frac{1 + \cos 2t}{2} dt = \left[\frac{t}{2} + \frac{\sin 2t}{4}\right]_0^{2\pi}$$

$$= \pi + \frac{1}{2}\big(\sin 4\pi - \sin 0\big) = \pi$$

—

If F is a field on some domain D,

$$F = \nabla f \quad \text{for some scalar } f,$$

then f is known as a potential function.

$$dr = \hat{i}dx + \hat{j}dy + \hat{k}dz, \quad \nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$
$$F \cdot dr = \nabla f \cdot dr = df$$
$$\int_A^B F \cdot dr = \int_A^B df = f(B) - f(A)$$

Hence, over a closed path:

$$\oint F \cdot dr = 0$$

i.e., the work done over a closed path is zero.

—

Let

$$F = M(x,y,z)\hat{i} + N(x,y,z)\hat{j} + P(x,y,z)\hat{k}.$$

F is conservative if and only if:

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

$$F = \hat{i}\frac{\partial f}{\partial x} + \hat{j}\frac{\partial f}{\partial y} + \hat{k}\frac{\partial f}{\partial z}$$

Now,

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial N}{\partial z}, \quad M = \frac{\partial f}{\partial x}$$
$$\frac{\partial M}{\partial z} = \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial z \partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial P}{\partial x}, \quad P = \frac{\partial f}{\partial z}$$
$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial M}{\partial y}, \quad N = \frac{\partial f}{\partial y}$$

—

Example: Conservative Field Check

$$F = e^{xz}(xy+z)\hat{i} + (xze^{xz} - e^{xz}\sin y)\hat{j} + (xy+z)e^{xz}\hat{k}$$
$$M = e^{xz}(xy+z), \quad N = xze^{xz} - e^{xz}\sin y, \quad P = (xy+z)e^{xz}$$

Check if F is conservative:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\big((xy+z)e^{xz}\big) = xe^{xz}$$
$$\frac{\partial N}{\partial z} = \frac{\partial}{\partial z}\big(xze^{xz} - e^{xz}\sin y\big) = xe^{xz}(z+1)$$
$$\frac{\partial M}{\partial z} = \frac{\partial}{\partial z}\big(e^{xz}(xy+z)\big) = xe^{xz}(xy+z) + e^{xz}$$
$$\frac{\partial P}{\partial x} = \cdots$$

Hence F is conservative. Potential function is obtained by integrating step by step:

$$f = e^{xz}\cos y + xyz + \frac{z^2}{2} + c$$

—

Newton's Law of Gravitation

$$F = \frac{GmM}{R^2}\hat{R} = \frac{GmM}{|R|^3}\vec{R}$$

$$= \frac{GmM(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$$

—

Rectangle Method for Circulation

Consider rectangle with vertices (x, y), $(x + \Delta x, y)$, $(x, y + \Delta y)$, $(x + \Delta x, y + \Delta y)$.

$$F(x, y) = M(x, y)\hat{i} + N(x, y)\hat{j}$$

Top side:

$$F(x, y + \Delta y) \cdot (j\Delta y) = N(x, y + \Delta y)\Delta x$$

Bottom side:

$$F(x, y) \cdot (-j\Delta y) = -N(x, y)\Delta x$$

Right side:

$$F(x + \Delta x, y) \cdot (i\Delta x) = M(x + \Delta x, y)\Delta y$$

Left side:

$$F(x, y) \cdot (-i\Delta x) = -M(x, y)\Delta y$$

Combining top and bottom:

$$\big(N(x, y + \Delta y) - N(x, y)\big)\Delta x \approx \frac{\partial N}{\partial y}\Delta x\Delta y$$

Combining left and right:

$$\big(M(x + \Delta x, y) - M(x, y)\big)\Delta y \approx \frac{\partial M}{\partial x}\Delta x\Delta y$$

Therefore,

$$\nabla \cdot F = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$$

Divergence Theorem in 2D (Green's Theorem - Divergence Form)

$$\oint \vec{F} \cdot \hat{n}\, ds = \oint (M\, dx - N\, dy)$$

$$\Updownarrow$$

$$\iint_{\text{Region}} \nabla \cdot \vec{F}\, dS = \iint_{\text{Region}} \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dx\, dy$$

Line Integrals over Piecewise Paths

Q7.

Figure shows path from origin to $(1,1,1)$ the union of line segments C_1, C_2.
Integrate $f(x,y,z) = x - 3y + z$ over $C_1 \cup C_2$:

$$\int_{C_1 \cup C_2} f(x,y,z)\, ds$$

Let $R(t) = t\hat{i} + t\hat{j}$ for $0 \le t \le 1$ (path C_1).

$$\vec{N}(t) = \frac{dR(t)}{dt}, \quad V(t) = \left|\frac{dR}{dt}\right|$$

So,

$$\int_{C_1 \cup C_2} f(x,y,z)\, ds = \int_{C_1} f(x(t),y(t),z(t))V(t)\, dt + \int_{C_2} f(x(t),y(t),z(t))V(t)\, dt$$

For C_1: $R(t) = t\hat{i} + t\hat{j}$, $0 \le t \le 1$, $V(t) = \sqrt{2}$

$$\int_{C_1} f(x,y,z)\, ds = \int_0^1 f(t,t,0)\sqrt{2}\, dt$$

For C_2: $R(t) = \hat{i} + t\hat{j} + t\hat{k}$, $0 \le t \le 1$, $V(t) = \sqrt{2}$

$$\int_{C_2} f(x,y,z)\, ds = \int_0^1 f(1,t,t)\, dt$$

Hence,

$$= \int_0^1 (t-3t)\sqrt{2}\, dt + \int_0^1 (1-3t+t)\, dt$$

$$= \frac{\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}\left[\frac{t^2}{2}\right]_0^1 + \left[t - t^2\right]_0^1$$

$$= \frac{\sqrt{2}}{2} - \sqrt{2}\cdot\frac{1}{2} - (1-0) = -2 - \frac{\sqrt{2}}{2} - \frac{3}{2}$$

—

Q8.

Integrate $f(x,y,z) = x + \sqrt{y} - z^2$ over the path from $(0,0,0)$ to $(1,1,1)$.
Path shown: C_1, C_2, C_3

$$C_1 : r(t) = t\hat{i}, \quad 0 \le t \le 1, \quad |r'(t)| = 1$$
$$C_2 : r(t) = \hat{i} + t\hat{j}, \quad 0 \le t \le 1, \quad |r'(t)| = 1$$
$$C_3 : r(t) = \hat{i} + \hat{j} + t\hat{k}, \quad 0 \le t \le 1, \quad |r'(t)| = 1$$

Thus,

$$\int f(x,y,z)\, ds = \int_{C_1} f(x,0,0)\, dt + \int_{C_2} f(1,y,0)\, dt + \int_{C_3} f(1,1,z)\, dt$$

$$= \int_0^1 t\, dt + \int_0^1 \left(1 + \sqrt{t}\right)\, dt + \int_0^1 (1 + 1 - t^2)\, dt$$

$$= \frac{1}{2} + \left[t + \frac{2}{3}t^{3/2}\right]_0^1 + \left[2t - \frac{t^3}{3}\right]_0^1$$

$$= \frac{1}{2} + \frac{5}{3} + \frac{5}{3} = \frac{1}{2} + \frac{10}{3} = \frac{1}{2} + \frac{10}{3} = \frac{23}{6}$$

Flux Across a Plane Curve

Flux of $\vec{F}$ across C:

$$\int_C \vec{F} \cdot \hat{n}\, ds$$

Let

$$\vec{r} = x\hat{i} + y\hat{j}, \quad \frac{d\vec{r}}{ds} = \hat{T}, \quad \hat{n} = \hat{T} \times \hat{k}$$

$$\hat{T} = \frac{d\vec{r}}{ds} = \hat{i}\frac{dx}{ds} + \hat{j}\frac{dy}{ds}$$

So,

$$\hat{n} = \left(\hat{i}\frac{dx}{ds} + \hat{j}\frac{dy}{ds}\right) \times \hat{k} = \hat{i}\frac{dy}{ds} - \hat{j}\frac{dx}{ds}$$

—
Let

$$\vec{F} = M(x,y)\hat{i} + N(x,y)\hat{j}$$

Then,

$$\int_C \vec{F} \cdot \hat{n}\, ds = \int_C \left(M\hat{i} + N\hat{j}\right) \cdot \left(\hat{i}\frac{dy}{ds} - \hat{j}\frac{dx}{ds}\right) ds$$

$$= \int_C \left(M\frac{dy}{ds} - N\frac{dx}{ds}\right) ds$$

$$\therefore \quad \int_C \vec{F} \cdot \hat{n}\, ds = \int_C \left(M\, dy - N\, dx\right)$$

Arc Length Along a Curve

Let

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

The arc length from $t = a$ to $t = b$:

$$L = \int_a^b \left|\frac{d\vec{r}}{dt}\right| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

Or, $L = \int_a^b |\vec{v}|\, dt$, where $\vec{v} = \frac{d\vec{r}}{dt}$.

Length of a Helix

Given $\vec{r}(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\hat{k}$, $0 \leq t \leq 2\pi$,

$$\frac{d\vec{r}}{dt} = -\sin t\,\hat{i} + \cos t\,\hat{j} + \hat{k} \implies |\vec{v}| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{1+1} = \sqrt{2}$$

$$L = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$

Tangent Vector to a Curve

The unit tangent vector:

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{d\vec{r}}{dt} \bigg/ \left|\frac{d\vec{r}}{dt}\right|$$

For parametrized curves $\vec{r}(t)$,

$$\vec{T} = \frac{d\vec{r}}{ds}$$

where s is the arc length parameter.

Curvature k

Curvature is defined by:

$$k = \left|\frac{d\vec{T}}{ds}\right|$$

In terms of t,

$$k = \left|\frac{d\vec{T}}{dt} \bigg/ \frac{ds}{dt}\right| = \frac{\left|\frac{d\vec{T}}{dt}\right|}{|\vec{v}|}$$

For a helix $\vec{r}(t) = a\cos t\,\hat{i} + a\sin t\,\hat{j} + bt\,\hat{k}$,

$$\vec{v} = -a\sin t\,\hat{i} + a\cos t\,\hat{j} + b\hat{k}, \quad |\vec{v}| = \sqrt{a^2+b^2}$$

$$\frac{d\vec{v}}{dt} = -a\cos t\,\hat{i} - a\sin t\,\hat{j}$$

$$k = \frac{a}{a^2+b^2}$$

Parametric Equations (Line and Plane)

A line in space through $P(x_0, y_0, z_0)$ parallel to $\vec{v} = A\hat{i} + B\hat{j} + C\hat{k}$:

$$x = x_0 + tA$$
$$y = y_0 + tB$$
$$z = z_0 + tC$$

A plane through $P(x_0, y_0, z_0)$ normal to $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$:

$$A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$$

Partial Derivatives and Continuity

For $f(x, y)$, the partial derivatives at (x_0, y_0):

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h}$$

Continuity at (x_0, y_0):

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

Extrema of $f(x, y)$

Find local extremum by solving:

$$f_x = 0, \quad f_y = 0$$

Second derivative test:

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

If $D > 0$, $f_{xx} > 0$: local min.
If $D > 0$, $f_{xx} < 0$: local max.
If $D < 0$: saddle point.

Green's Theorem

$$\oint_C M(x, y)\, dx + N(x, y)\, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

Conservative Vector Field

The vector field $\vec{F} = M(x, y, z)\hat{i} + N(x, y, z)\hat{j} + P(x, y, z)\hat{k}$ is conservative if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$$

Arc Length for Circle and Ellipse

For a circle ($x = r\cos t$, $y = r\sin t$, $0 \le t \le 2\pi$),

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{2\pi} r\, dt = 2\pi r$$

For an ellipse ($x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$),

$$L = \int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}\, dt$$

General Results

Chain rule for composite functions:

$$w = f(g(r, s), h(r, s), k(r, s))$$
$$\frac{dw}{dr} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}$$
$$\frac{dw}{ds} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s}$$

Sample Vector Calculus Identities

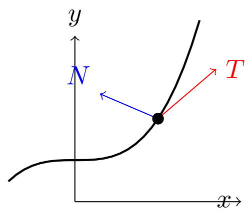
Product rule for dot product:

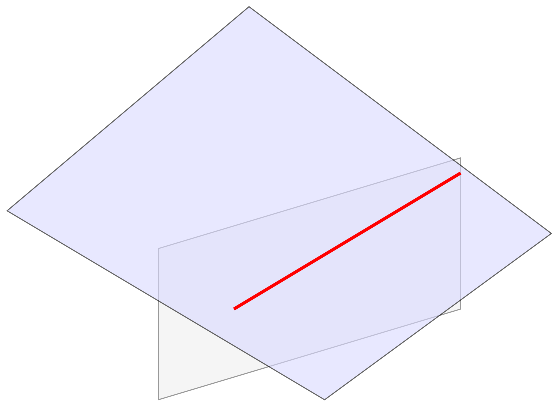
$$\frac{d}{dt}\left(\vec{U} \cdot \vec{V}\right) = \frac{d\vec{U}}{dt} \cdot \vec{V} + \vec{U} \cdot \frac{d\vec{V}}{dt}$$

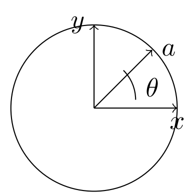
If $|\vec{U}|$ is constant,

$$\vec{U} \cdot \frac{d\vec{U}}{dt} = 0$$

Diagrams







Circulation around closed loop and Green's Theorem

Let the region R be a rectangular area. Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\, dy$$

Tangent vector:

$$\vec{T} = \hat{i}\frac{dx}{ds} + \hat{j}\frac{dy}{ds}$$

$$\vec{F} = M\hat{i} + N\hat{j}$$

$$\vec{F} \cdot \vec{T}\, ds = \left(M\frac{dx}{ds} + N\frac{dy}{ds} \right) ds$$

Verification of Green's Theorem for $\vec{F}(x, y) = (x - y)\hat{i} + x\hat{j}$

Region R bounded by unit circle $x^2 + y^2 = 1$

Parametrize:

$$x = \cos t, \quad y = \sin t, \quad dx = -\sin t\, dt, \quad dy = \cos t\, dt, \quad 0 \leq t \leq 2\pi$$

$M = x - y$, $N = x$

$$\frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1$$

$\oint_C M dy - N dx$:

$$\int_0^{2\pi} (\cos t - \sin t) \cos t \, dt - \int_0^{2\pi} \cos t(-\sin t) dt$$

$$= \int_0^{2\pi} \cos^2 t \, dt - \int_0^{2\pi} \sin t \, \cos t \, dt + \int_0^{2\pi} \sin t \cos t \, dt$$

$$= \int_0^{2\pi} \cos^2 t \, dt$$

$$= \frac{1}{2} \int_0^{2\pi} (1 + \cos 2t) dt = \pi$$

$\iint_R dx \, dy = \pi$

$$\oint_C M dx + N dy$$

$$= \int_0^{2\pi} (\cos t - \sin t)(-\sin t) dt + \int_0^{2\pi} \cos t \cos t \, dt$$

$$= -\int_0^{2\pi} \cos t \sin t \, dt + \int_0^{2\pi} \sin^2 t \, dt + \int_0^{2\pi} \cos^2 t \, dt$$

$$= \int_0^{2\pi} (1 - \cos^2 t) \, dt + \int_0^{2\pi} \cos^2 t \, dt$$

$$= \int_0^{2\pi} 1 \, dt = 2\pi$$

Evaluate $\oint_C x \, dy - y^2 \, dx$ where C is the square with vertices in first quadrant, $x = y = 1$

Green's theorem:

$$\oint_C x \, dy - y^2 \, dx = \iint_R \left(\frac{\partial x}{\partial x} - \frac{\partial y^2}{\partial y} \right) dx \, dy$$

$$\frac{\partial x}{\partial x} = 1, \quad \frac{\partial y^2}{\partial y} = 2y$$

$$\iint_R (1 - 2y) dx \, dy$$

Work examples

$$M = x, \, N = -y^2$$

$$N = xy$$

$$\iint_R y \, dx \, dy - \iint_R 2y \, dx \, dy = \frac{3}{2}$$

$$M = xy, \, N = xy$$

$$\iint_R y - (-2y) dx \, dy = \frac{3}{2}$$

Flux and Conservative Vector Fields

1. Calculate outward flux of field $\vec{F}(x,y) = x\hat{i} + y\hat{j}$ across square bounded by the lines $x = 1, y = 1, x = -1, y = -1$:

Let $M = x$, $N = y$.

$$\iint \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\, dy = \iint (0-0) dx\, dy = 0$$

But if integrating

$$\iint \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dx\, dy = \iint (1+1) dx\, dy = 2 \iint dx\, dy$$

Area of square is 4, so total flux is 8.

$$\iint (1+1) dx\, dy = 2 \int_{-1}^{1} \int_{-1}^{1} dy\, dx = 2 \cdot 2 \cdot 2 = 8$$

2. Find work done by conservative field $\vec{F} = y^2\hat{i} + xz\hat{j} + xy\hat{k} = \nabla f(x,y,z)$ along any smooth curve joining the point A $(-1, 5, 9)$ to $B = (1, 6, 12)$.

$$F = \nabla f$$

$$df = \vec{\nabla} f \cdot d\vec{r}$$

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$

Example calculation:

$$f(B) = (6)(12) + (1)(6) + (1)(12) = 72 + 6 + 12 = 90 f(A) = (5)(9) + (-1)(5) + (-1)(9) = 45 - 5 - 9 = 31 W = f(B) - f(A) = 90 - 31$$

Another method:

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\vec{F} \cdot d\vec{r} = (y^2\hat{i} + xz\hat{j} + xy\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = y^2 dx + xz dy + xy dz$$

$$\int_C y^2 dx + xz dy + xy dz$$

Differential Forms and Conservative Fields

The force $M(x,y,z)\, dx + N(x,y,z)\, dy + P(x,y,z)\, dz$ is called a differential form.

A differential form is exact on domain D in space if:

$$M dx + N dy + P dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

The test for exactness:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$$

This is equivalent to saying that the field $\vec{F} = M(x,y,z)\hat{i} + N(x,y,z)\hat{j} + P(x,y,z)\hat{k}$ is conservative.

Date: 18/09/2025

Arc Length Along a Curve

Let

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

Arc length from $t = a$ to $t = b$:

$$\text{length} = k = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

Or, more generally:

$$L = \int_a^b \left|\frac{d\vec{r}}{dt}\right| dt = \int_a^b |\vec{v}| dt$$

$$S(t) = \int_0^t |\vec{v}|\, d\tau$$

Helix Example

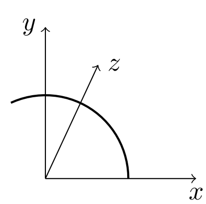
Find length of one turn of helix:

$$\vec{r}(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\,\hat{k}, \quad 0 \le t \le 2\pi$$

$$\frac{d\vec{r}}{dt} = -\sin t\,\hat{i} + \cos t\,\hat{j} + \hat{k}$$

$$|\vec{v}| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{1+1} = \sqrt{2}$$

$$L = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$



Unit Tangent Vector for a Curve

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{d\vec{r}}{dt}\bigg|\frac{d\vec{r}}{dt}\bigg|$$

Or, if parameterized by arc length s:

$$\text{Unit tangent vector: } \vec{T} = \frac{d\vec{r}}{ds}$$

If $\vec{v} = a\hat{i} + b\hat{j}$ then $|\vec{v}| = \sqrt{a^2 + b^2}$

Find tangent to right hand hyperbola branch at point $(\sqrt{2}, 1)$, $t = \frac{\pi}{4}$

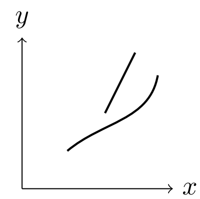
Given:

$$x = \sec t, \quad y = \tan t$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\sec^2 t}{\sec t \tan t}$$

At $t = \frac{\pi}{4}$:

$$= \sqrt{2}$$



Curvature for Helix

Let

$$\vec{r}(t) = a\cos t\,\hat{i} + a\sin t\,\hat{j} + bt\,\hat{k}$$

Unit tangent vector:

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{d\vec{r}(t)}{dt} \Big/ \left|\frac{d\vec{r}(t)}{dt}\right| = \frac{-a\sin t\,\hat{i} + a\cos t\,\hat{j} + b\,\hat{k}}{\sqrt{a^2+b^2}}$$

Curvature:

$$k = \left|\frac{d\vec{T}}{ds}\right| = \left|\frac{\frac{d\vec{T}}{dt}}{|\vec{v}|}\right|$$

Calculate derivative:

$$\frac{d\vec{T}}{dt} = \frac{-a\cos t\,\hat{i} - a\sin t\,\hat{j}}{\sqrt{a^2+b^2}}$$

So,

$$k = \frac{\left|-a\cos t\,\hat{i} - a\sin t\,\hat{j}\right|}{a^2+b^2}$$
$$= \frac{a}{a^2+b^2}$$

Partial Derivatives and Continuity

We say that $f(x, y)$ approaches L as $(x, y) \to (x_0, y_0)$ if

$$\lim_{x\to x_0,\ y\to y_0} f(x,y) = L$$

For every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f:

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

Function f is continuous at (x_0, y_0) if this condition is satisfied.

Tangent to Curve

Consider the point-slope form:

$$y - y_0 = m(x - x_0)$$

If $y_0 = 1$, $x_0 = \sqrt{2}$ and $m = \sqrt{2}$,

$$y - 1 = \sqrt{2}(x - \sqrt{2}) \implies y = \sqrt{2}x - 2 \implies y = \sqrt{2}x - 1$$

Arc Length for Circle

For circle with parametric equations:

$$x = r\cos t, \quad y = r\sin t, \quad 0 \le t \le 2\pi$$

$$\frac{dx}{dt} = -r\sin t, \quad \frac{dy}{dt} = r\cos t$$

Arc length:

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{2\pi} r\, dt = 2\pi r$$

Arc Length for Ellipse

For ellipse:

$$x = a\cos t, \quad y = b\sin t, \quad 0 \le t \le 2\pi$$

$$\frac{dx}{dt} = -a\sin t, \quad \frac{dy}{dt} = b\cos t$$

Arc length:

$$L = \int_0^{2\pi} \sqrt{a^2\sin^2 t + b^2\cos^2 t} dt$$

$$= \int_0^{2\pi} \sqrt{a^2(1 - \cos^2 t) + b^2\cos^2 t} dt = \int_0^{2\pi} \sqrt{a^2 - (a^2 - b^2)\cos^2 t} dt$$

or

$$L = a\int_0^{2\pi} \sqrt{1 - e^2\cos^2 t} dt, \quad \text{where } e^2 = \frac{a^2 - b^2}{a^2}$$

Curvature

If T is the unit tangent vector of a smooth curve, the curvature is

$$k = \left|\frac{dT}{ds}\right|$$

For a circle of radius r:

$$k = \frac{1}{r}$$

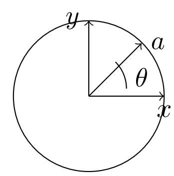
Parametric Circle Example

Let $s = a\theta$,

$$\vec{r}(\theta) = a\cos\theta\,\hat{i} + a\sin\theta\,\hat{j}$$

$$T(\theta) = \frac{d\vec{r}}{ds} = \frac{d\vec{r}/d\theta}{ds/d\theta} = \frac{-a\sin\theta\,\hat{i} + a\cos\theta\,\hat{j}}{a} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

$$K = \left|\frac{dT}{ds}\right| = \left|-\frac{1}{a}\cos\left(\frac{s}{a}\right)\hat{i} - \frac{1}{a}\sin\left(\frac{s}{a}\right)\hat{j}\right| = \frac{1}{a}$$

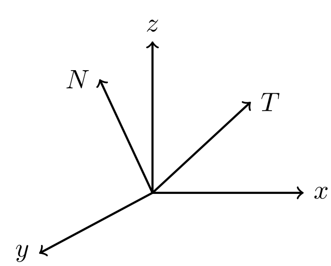


Curvature and Tangent/Normal Vectors

$$k = \left|\frac{dT}{ds}\right| = \frac{1}{a}$$

For $y = mx + c$, curvature at any point is zero.
At a point where $k \neq 0$, principal unit normal vector:

$$\mathbf{N} = \frac{1}{k}\frac{dT}{ds} = \frac{\frac{dT}{dt}}{\left|\frac{dT}{dt}\right|}$$



Example: Find T and N for planar motion

Given $\vec{r}(t) = \cos 2t\,\hat{i} + \sin 2t\,\hat{j}$

$$T = \frac{d\vec{r}(t)}{dt}\left|\frac{d\vec{r}(t)}{dt}\right|$$

$$\frac{d\vec{r}}{dt} = -2\sin 2t\,\hat{i} + 2\cos 2t\,\hat{j}$$

$$|\frac{d\vec{r}}{dt}| = \sqrt{4\sin^2 2t + 4\cos^2 2t} = 2$$

$$T = \frac{-2\sin 2t\,\hat{i} + 2\cos 2t\,\hat{j}}{2} = -\sin 2t\,\hat{i} + \cos 2t\,\hat{j}$$

$$k = \left|\frac{dT}{dt}\right|/|\vec{v}| = \frac{|-2\cos 2t\,\hat{i} - 2\sin 2t\,\hat{j}|}{2} = 1$$

$$N = \frac{dT/dt}{|dT/dt|} = \frac{-2\cos 2t\,\hat{i} - 2\sin 2t\,\hat{j}}{2} = -\cos 2t\,\hat{i} - \sin 2t\,\hat{j}$$

Find a vector perpendicular to the plane of $P(1,-1,0)$, $Q(3,1,-1)$, $R(7,1,2)$

Let

$$\overrightarrow{PQ} = 2\hat{i} + 2\hat{j} - \hat{k}, \quad \overrightarrow{PR} = 6\hat{i} + 2\hat{j} + 2\hat{k}$$

Cross product:

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 6 & 2 & 2 \end{vmatrix} = (2 \cdot 2 - (-1) \cdot 2)\hat{i} - (2 \cdot 2 - (-1) \cdot 6)\hat{j} + (2 \cdot 2 - 2 \cdot 6)\hat{k} = 6\hat{i} - 6\hat{j} + 12\hat{k}$$

Unit vector:

$$\vec{U} = \frac{6\hat{i} - 6\hat{j} + 12\hat{k}}{\sqrt{6^2 + (-6)^2 + 12^2}} = \frac{6\hat{i} - 6\hat{j} + 12\hat{k}}{\sqrt{36 + 36 + 144}} = \frac{6\hat{i} - 6\hat{j} + 12\hat{k}}{\sqrt{216}} = \frac{6\hat{i} - 6\hat{j} + 12\hat{k}}{14.7}$$

Box Volume and Scalar Triple Product

$$\text{Area of box } = |\vec{A} \times \vec{B}|; \quad \text{Height } = |C|\cos\theta$$

$$\text{Volume} = \text{Area} \times \text{Height} = |\vec{A} \times \vec{B}||C|\cos\theta$$

$$\text{For rectangular parallelopiped: } |\vec{A} \times \vec{B}||C|\cos\theta = |\vec{C} \cdot (\vec{A} \times \vec{B})|$$

General formula:

$$\vec{A} = a_i\hat{i} + a_j\hat{j} + a_k\hat{k}\vec{B} = b_i\hat{i} + b_j\hat{j} + b_k\hat{k}\vec{C} = c_i\hat{i} + c_j\hat{j} + c_k\hat{k}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Find volume determined by $\vec{A} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{C} = 7\hat{j} - 4\hat{k}$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 0 & 7 & -4 \end{vmatrix} = -23$$

$$\boxed{|\vec{A} \cdot (\vec{B} \times \vec{C})| = 23}$$

Tangent and Normal Vectors; Sphere Geometry

$$\oint \vec{F} \cdot d\vec{s} = \oint (M dx - N dy)$$

$$\iint \vec{\nabla} \cdot \vec{F} ds = \iint \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx\, dy$$

Find unit vectors tangent and normal to $y = x^2 + \frac{1}{2}$ at $(1, 1.5)$
If $\vec{v} = a_1\hat{i} + b_1\hat{j}$,

$$\left.\frac{dy}{dx}\right|_{(1,1.5)} = 2x = 2 \implies \text{slope} = 2$$

Tangent vector $\vec{v} = \hat{i} + 2\hat{j}$
Unit tangent:

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{1\hat{i} + 2\hat{j}}{\sqrt{1^2 + 2^2}} = \frac{1\hat{i} + 2\hat{j}}{\sqrt{5}}$$

Unit normal is perpendicular; if $\vec{u} = a\hat{i} + b\hat{j}$, then one normal is $-b\hat{i} + a\hat{j}$
So,

$$\vec{n} = \frac{-2\hat{i} + 1\hat{j}}{\sqrt{5}}$$

Vector between Points and Sphere Center/Radius

Distance from $P(x_1, y_1, z_1)$ to $P(x_2, y_2, z_2)$:

$$|\overrightarrow{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Find the unit vector in the direction of $\overrightarrow{P_1P_2}$:

$$\frac{\overrightarrow{P_1P_2}}{|\overrightarrow{P_1P_2}|}$$

Equation of a sphere:

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

Center $= (x_0, y_0, z_0)$, Radius $= a$
Example:

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0$$

Complete the square:

$$(x + \frac{3}{2})^2 + y^2 + (z - 2)^2 = (\frac{\sqrt{21}}{2})^2$$

So center: $(-\frac{3}{2}, 0, 2)$, radius: $\frac{\sqrt{21}}{2}$

Lines and Line Segments in Space

Suppose a line in space passing through $P(x_1, y_1, z_1)$ is parallel to vector $\vec{v}$:

$$\vec{v} = A\hat{i} + B\hat{j} + C\hat{k}$$

Equation for the line:

$$x = x_0 + tA, \quad y = y_0 + tB, \quad z = z_0 + tC$$

Example 1: Line through $(-2, 0, 4)$ parallel to $\vec{v} = 2\hat{i} + 4\hat{j} - 2\hat{k}$:

$$x = -2 + 2t,\ y = 0 + 4t,\ z = 4 - 2t$$

Example 2: Line through $P(-3, 2, -3)$ and $Q(1, 1, 4)$:

$$\overrightarrow{PQ} = 4\hat{i} - 1\hat{j} + 7\hat{k}$$

$$x = -3 + 4t,\ y = 2 - t,\ z = -3 + 7t$$

Plane through $P_0(x_0, y_0, z_0)$:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Example: Plane through $(-3, 0, 7)$ normal to $5\hat{i} + 2\hat{j} - \hat{k}$:

$$5(x + 3) + 2y - 1(z - 7) = 0 \Rightarrow 5x + 2y - z = -28$$

Time Derivatives and Local Extrema

$$\frac{d\vec{U}}{dt} = \frac{dU_1}{dt}\hat{i} + \frac{dU_2}{dt}\hat{j} + \frac{dU_3}{dt}\hat{k}$$

If $\vec{U}$ is a differentiable vector of constant length, then

$$\vec{U} \cdot \frac{d\vec{U}}{dt} = 0$$

If $|\vec{U}|^2$ is constant:

$$\frac{d}{dt}(\vec{U} \cdot \vec{U}) = 2\vec{U} \cdot \frac{d\vec{U}}{dt} = 0$$

Example: $\vec{U}(t) = \sin t\hat{i} + \cos t\hat{j} + \sqrt{3}\hat{k}$

$$\frac{d\vec{U}}{dt} = \cos t\hat{i} - \sin t\hat{j}$$

$$|\vec{U}(t)| = \sqrt{\sin^2 t + \cos^2 t + 3} = \sqrt{1+3} = 2$$

Derivative and Product Rule

Product rule for dot product:

$$\frac{d}{dt}(\vec{U} \cdot \vec{V}) = \frac{dU_1}{dt}V_1 + U_1\frac{dV_1}{dt} + \frac{dU_2}{dt}V_2 + U_2\frac{dV_2}{dt} + \frac{dU_3}{dt}V_3 + U_3\frac{dV_3}{dt}$$

Or in vector form,

$$\frac{d(\vec{U} \cdot \vec{V})}{dt} = \vec{U} \cdot \frac{d\vec{V}}{dt} + \frac{d\vec{U}}{dt} \cdot \vec{V}$$

For limit of vector as $t \to t_0$:

$$\lim_{t \to t_0} \vec{r}(t) = \vec{L}$$

where $\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$.

General Leibniz rule for product:

$$\frac{d}{dt}(a(t)b(t)) = a'(t)b(t) + a(t)b'(t)$$

Intersection and Angle between Planes

1. Find point where line $x = \frac{3}{5} + 2t$, $y = -2t$, $z = 1 + t$ intersects plane $3x + 2y + 6z = 6$:

$$3\left(\frac{3}{5} + 2t\right) + 2(-2t) + 6(1+t) = 6$$
$$\frac{9}{5} + 6t - 4t + 6 + 6t = 6$$
$$\frac{9}{5} + 2t + 6 + 6t = 6$$
$$8t + \frac{9}{5} + 6 = 6$$
$$8t = 6 - 6 - \frac{9}{5} = -\frac{9}{5}$$
$$t = -\frac{9}{40}$$

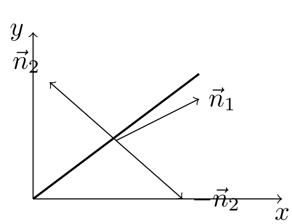
2. Find angle between planes $3x - 6y - 2 = 15$ and $2x + y - 22 = 5$:
Normals:

$$\vec{n}_1 = 3\hat{i} - 6\hat{j} - 2\hat{k}$$
$$\vec{n}_2 = 2\hat{i} + \hat{j} - \hat{k}$$
$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$
$$\vec{n}_1 \cdot \vec{n}_2 = 3 \cdot 2 + (-6) \cdot 1 + (-2) \cdot (-1) = 6 - 6 + 2 = 2$$
$$|\vec{n}_1| = \sqrt{3^2 + (-6)^2 + (-2)^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7$$
$$|\vec{n}_2| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$
$$\cos\theta = \frac{2}{7\sqrt{6}}$$

3. Find vector parallel to line of intersection of planes:

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -6 & -2 \\ 2 & 1 & -1 \end{vmatrix}$$

Slope, Tangent, and Unit Vectors



Slope at $(1,1)$ is
$$\left.\frac{dy}{dx}\right|_{(1,1)} = \frac{3}{2}$$

Given $\vec{V} = 2\hat{i} + 3\hat{j}$, its magnitude is:
$$|\vec{V}| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

Normalized (unit) vector:
$$\frac{\vec{V}}{|\vec{V}|} = \frac{2\hat{i} + 3\hat{j}}{\sqrt{13}}$$

Negative unit vector:
$$-\frac{\vec{V}}{|\vec{V}|} = \frac{-2\hat{i} - 3\hat{j}}{\sqrt{13}}$$

Differential Forms and Conservativeness

The form $M(x,y,z)\,dx + N(x,y,z)\,dy + P(x,y,z)\,dz$ is called a differential form.

A differential form is **exact** on domain D in space if:
$$M\,dx + N\,dy + P\,dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

The test for exactness: Differential form $M\,dx + N\,dy + P\,dz$ is exact if and only if:
$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$$

This is equivalent to saying that the field $\vec{F} = M(x,y,z)\hat{i} + N(x,y,z)\hat{j} + P(x,y,z)\hat{k}$ is conservative.

Vector Calculations in the Plane

$$|\overrightarrow{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\hat{n} = \frac{\overrightarrow{P_1P_2}}{|\overrightarrow{P_1P_2}|}$$

If $v = a\hat{i} + b\hat{j}$, slope $= \frac{b}{a}$.

Find Unit Tangent and Normal (Example)

For $y = \frac{x^3}{2} + \frac{1}{2}$ at $(1,1)$,
$$\frac{dy}{dx} = \frac{3x^2}{2}, \quad \text{at } x = 1 \implies \frac{3}{2}$$

Tangent vector:
$$\vec{v} = \hat{i} + \frac{3}{2}\hat{j}$$

Unit tangent:
$$\frac{\vec{v}}{|\vec{v}|} = \frac{\hat{i} + \frac{3}{2}\hat{j}}{\sqrt{1^2 + (3/2)^2}} = \frac{\hat{i} + \frac{3}{2}\hat{j}}{\sqrt{1 + 9/4}} = \frac{\hat{i} + \frac{3}{2}\hat{j}}{\sqrt{13/4}} = \frac{2}{\sqrt{13}}(\hat{i} + \frac{3}{2}\hat{j})$$

Unit normal (perpendicular direction):
$$\frac{-\frac{3}{2}\hat{i} + \hat{j}}{\sqrt{13/4}} = \frac{-3\hat{i} + 2\hat{j}}{\sqrt{13}}$$

Green's Theorem and Flux

Green's theorem relates the circulation around a simple closed curve C bounding region R to a double integral over R:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

where $\mathbf{F} = M(x,y)\mathbf{i} + N(x,y)\mathbf{j}$.

Verification Example:

Let $\mathbf{F} = (x-y)\mathbf{i} + x\mathbf{j}$ and C be the unit circle $x^2 + y^2 = 1$ parametrized by:

$$x = \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi$$

Then,

$$M = x - y, \quad N = x$$

Calculate partial derivatives:

$$\frac{\partial N}{\partial x} = 1, \quad \frac{\partial M}{\partial y} = -1$$

Evaluate the right-hand side of Green's theorem:

$$\iint_R (1-(-1))dA = \iint_R 2\,dA = 2 \times \text{Area of unit circle} = 2\pi$$

Now, the line integral around C:

$$\oint_C M dy - N dx = \int_0^{2\pi} (x-y)dy - xdx = \pi \quad \text{(calculated in steps above)}$$

And the alternate circulation form value is 2π. Details are shown in the calculations above.

Flux Computation

Calculate flux of $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ over square bounded by $x = \pm 1$, $y = \pm 1$:

Using divergence theorem:

$$\iint_R \nabla \cdot \mathbf{F}\, dA = \iint_R \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} \right) dA = \iint_R (1+1)dA = 2 \times 4 = 8$$

Work Done by Conservative Vector Field

Given $\mathbf{F} = y^2\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ and points $A(-1,3,9), B(1,6,12)$:

Since $\mathbf{F} = \nabla f$, work done is

$$W = f(B) - f(A)$$

Find potential function $f(x,y,z) = xyz$:

$$f(B) = (1)(6)(12) = 72, \quad f(A) = (-1)(3)(9) = -27$$

Therefore,

$$W = 72 - (-27) = 99$$

Alternate:

$$\int_C y^2 dx + xzdy + xydz$$

Differential Forms

A differential form $Mdx + Ndy + Pdz$ is **exact** on domain D if:

$$Mdx + Ndy + Pdz = df = \nabla f \cdot dr$$

This holds iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$$

which means vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ is conservative.

Green's Theorem and Flux

Let R be a rectangular region and C its boundary. Green's Theorem states:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy,$$

where $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$.

Verification example: For $\mathbf{F} = (x - y)\mathbf{i} + x\mathbf{j}$ and C parameterized by $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, compute

$$M = x - y, \qquad N = x,$$
$$\frac{\partial N}{\partial x} = 1, \qquad \frac{\partial M}{\partial y} = -1.$$

The double integral evaluates to

$$\iint_R (1 - (-1))dxdy = \iint_R 2dxdy = 2 \times \pi = 2\pi.$$

The line integral calculations yield

$$\oint_C Mdy - Ndx = \pi,$$

and alternatively,

$$\oint_C Mdx + Ndy = 2\pi.$$

Flux over a Square

Calculate outward flux of $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ over square $[-1, 1] \times [-1, 1]$:

$$\text{Flux} = \iint_R \nabla \cdot \mathbf{F} dxdy = \iint_R \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} \right) dxdy = \iint_R 2dxdy = 2 \times 4 = 8.$$

Work Done by Conservative Vector Field

Given $\mathbf{F} = y^2\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ and path from $A(-1, 3, 9)$ to $B(1, 6, 12)$:

Since $\mathbf{F}$ is conservative with potential function $f(x, y, z) = xyz$, work done is

$$W = f(B) - f(A) = (1)(6)(12) - (-1)(3)(9) = 72 + 27 = 99.$$

Alternatively,

$$W = \int_C y^2 dx + xzdy + xydz.$$

Differential form and Conservativeness

A differential form

$$\omega = Mdx + Ndy + Pdz$$

is exact (or conservative) iff

$$\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}, \quad \frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}.$$

This implies $\omega = df$ for some scalar function f, and the associated vector field is conservative.

Arc Length of Parametric Curve

Given $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, arc length L from $t = a$ to $t = b$ is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

For helix

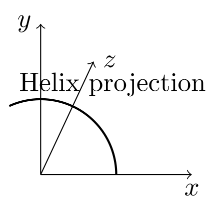
$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}, \quad 0 \le t \le 2\pi,$$

the magnitude of velocity is

$$|\mathbf{v}| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2},$$

thus, length of one turn is

$$L = \int_0^{2\pi} \sqrt{2}dt = 2\pi\sqrt{2}.$$



Unit Tangent Vector

$$\mathbf{T} = \frac{d\mathbf{r}/dt}{|d\mathbf{r}/dt|}.$$

Slope of curve at t:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}.$$

Example:
Curve $\mathbf{r}(t) = \sec t\mathbf{i} + \tan t\mathbf{j}$, compute slope at $t = \frac{\pi}{4}$,

$$\left.\frac{dy}{dx}\right|_{t=\pi/4} = \sqrt{2}.$$

Curvature of Helix

For

$$\mathbf{r}(t) = a\cos t\mathbf{i} + a\sin t\mathbf{j} + bt\mathbf{k},$$

curvature is

$$k = \frac{a}{a^2 + b^2}.$$

Partial derivatives and continuity

A function $f(x, y)$ approaches limit L at (x_0, y_0) if for every $\epsilon > 0$ there exists $\delta > 0$ such that when

$$\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta,$$

we have

$$|f(x, y) - L| < \epsilon.$$

This defines continuity at (x_0, y_0).

Circulation around closed loop and Green's Theorem

Let the region R be a rectangular area. Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\, dy$$

Tangent vector:

$$\vec{T} = \hat{\imath}\frac{dx}{ds} + \hat{\jmath}\frac{dy}{ds}$$
$$\vec{F} = M\hat{\imath} + N\hat{\jmath}$$
$$\vec{F} \cdot \vec{T}\, ds = \left(M\frac{dx}{ds} + N\frac{dy}{ds} \right) ds$$

Verification of Green's Theorem for $\vec{F}(x, y) = (x - y)\hat{\imath} + x\hat{\jmath}$

Region R bounded by unit circle $x^2 + y^2 = 1$
Parametrize:

$$x = \cos t, \quad y = \sin t, \quad dx = -\sin t\, dt, \quad dy = \cos t\, dt, \quad 0 \le t \le 2\pi$$

$M = x - y$, $N = x$

$$\frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1$$

Calculate line integral:

$$\oint_C M dy - N dx = \int_0^{2\pi} (\cos t - \sin t) \cos t\, dt - \int_0^{2\pi} \cos t(-\sin t) dt$$
$$= \int_0^{2\pi} \cos^2 t\, dt - \int_0^{2\pi} \sin t\, \cos t\, dt + \int_0^{2\pi} \sin t \cos t\, dt = \int_0^{2\pi} \cos^2 t\, dt$$
$$= \frac{1}{2} \int_0^{2\pi} (1 + \cos 2t) dt = \pi$$

Area integral:

$$\iint_R dx\, dy = \pi$$

Alternative circulation:

$$\oint_C M dx + N dy = \int_0^{2\pi} (\cos t - \sin t)(-\sin t) dt + \int_0^{2\pi} \cos t \cos t\, dt$$
$$= -\int_0^{2\pi} \cos t \sin t\, dt + \int_0^{2\pi} \sin^2 t\, dt + \int_0^{2\pi} \cos^2 t\, dt$$
$$= \int_0^{2\pi} (1 - \cos^2 t)\, dt + \int_0^{2\pi} \cos^2 t\, dt = \int_0^{2\pi} 1\, dt = 2\pi$$

Evaluate $\oint_C x\,dy - y^2\,dx$ where C is the square with vertices in first quadrant, $x = y = 1$

Green's theorem:

$$\oint_C x\,dy - y^2\,dx = \iint_R \left(\frac{\partial x}{\partial x} - \frac{\partial y^2}{\partial y}\right) dx\,dy$$

$$\frac{\partial x}{\partial x} = 1, \quad \frac{\partial y^2}{\partial y} = 2y$$

$$\iint_R (1-2y) dx\,dy$$

Work examples

$$M = x,\ N = -y^2$$

$$N = xy$$

$$\iint_R y\,dx\,dy - \iint_R 2y\,dx\,dy = \frac{3}{2}$$

$$M = xy,\ N = xy$$

$$\iint_R y - (-2y) dx\,dy = \frac{3}{2}$$

Flux and Conservative Vector Fields

1. Calculate outward flux of field $\vec{F}(x,y) = x\hat{\imath} + y\hat{\jmath}$ across square bounded by the lines $x = 1, y = 1, x = -1, y = -1$:

Let $M = x$, $N = y$.

$$\iint \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \iint (0-0) dx\,dy = 0$$

But if integrating

$$\iint \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y}\right) dx\,dy = \iint (1+1) dx\,dy = 2\iint dx\,dy$$

Area of square is 4, so total flux is 8.

$$\iint (1+1) dx\,dy = 2\int_{-1}^{1}\int_{-1}^{1} dy\,dx = 2\cdot 2\cdot 2 = 8$$

2. Find work done by conservative field $\vec{F} = y^2\hat{\imath} + xz\hat{\jmath} + xy\hat{k} = \nabla f(x,y,z)$ along any smooth curve joining the point $A = (-1,5,9)$ to $B = (1,6,12)$:

$$F = \nabla f, \quad df = \nabla f \cdot d\vec{r}$$

$$W = \int_C \vec{F}\cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$

Example calculation:

$$f(B) = (1)(6)(12) = 72, \quad f(A) = (-1)(5)(9) = -45$$

$$W = 72 - (-45) = 117$$

Another method:

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\vec{F} \cdot d\vec{r} = (y^2\hat{i} + xz\hat{j} + xy\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = y^2dx + xzdy + xydz$$

$$\int_C y^2dx + xzdy + xydz$$

Differential Forms and Conservative Fields

The force

$$M(x,y,z)dx + N(x,y,z)dy + P(x,y,z)dz$$

is called a differential form.

A differential form is exact in domain D if

$$Mdx + Ndy + Pdz = df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz.$$

The test for exactness is:

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

This is equivalent to the vector field

$$\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$$

being conservative.

Arc Length Along a Curve

Let

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

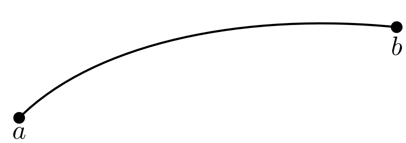
Arc length from $t = a$ to $t = b$:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt.$$

Or more generally,

$$L = \int_a^b \left|\frac{d\vec{r}}{dt}\right| dt.$$

$$S(t) = \int_0^t \left|\frac{d\vec{r}}{d\tau}\right| d\tau.$$



Helix Example

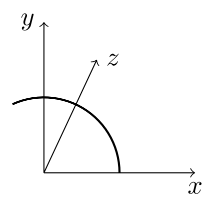
Find length of one turn of helix:

$$\vec{r}(t) = \cos t\hat{i} + \sin t\hat{j} + t\hat{k}, \quad 0 \le t \le 2\pi.$$

$$\frac{d\vec{r}}{dt} = -\sin t\hat{i} + \cos t\hat{j} + \hat{k}.$$

$$|\vec{v}| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}.$$

$$L = \int_0^{2\pi} \sqrt{2}dt = 2\pi\sqrt{2}.$$



Unit Tangent Vector for a Curve

$$\vec{T} = \frac{\frac{d\vec{r}}{dt}}{\left|\frac{d\vec{r}}{dt}\right|}$$

Or, if parameterized by arc length s:

$$\vec{T} = \frac{d\vec{r}}{ds}.$$

If $\vec{v} = a\hat{i} + b\hat{j}$, then $|\vec{v}| = \sqrt{a^2 + b^2}$.

Find Tangent to Right Hand Hyperbola Branch

Curve:

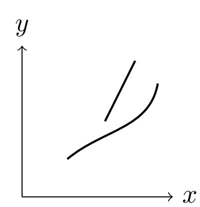
$$x = \sec t, \quad y = \tan t.$$

Slope:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\sec^2 t}{\sec t \tan t}.$$

At $t = \frac{\pi}{4}$:

$$\frac{dy}{dx} = \sqrt{2}.$$



Curvature for Helix

$$\vec{r}(t) = a\cos t\hat{i} + a\sin t\hat{j} + bt\hat{k}.$$

Unit tangent vector:

$$\vec{T} = \frac{d\vec{r}(t)/dt}{|d\vec{r}(t)/dt|} = \frac{-a\sin t\hat{i} + a\cos t\hat{j} + b\hat{k}}{\sqrt{a^2+b^2}}.$$

Curvature:

$$k = \left|\frac{d\vec{T}}{ds}\right| = \left|\frac{\frac{d\vec{T}}{dt}}{|d\vec{r}(t)/dt|}\right|.$$

Derivative:

$$\frac{d\vec{T}}{dt} = \frac{-a\cos t\hat{i} - a\sin t\hat{j}}{\sqrt{a^2+b^2}}.$$

Hence:

$$k = \frac{a}{a^2+b^2}.$$

Partial Derivatives and Continuity

Function $f(x,y)$ approaches L as $(x,y) \to (x_0,y_0)$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that:

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \implies |f(x,y) - L| < \epsilon.$$